OFFICIAL BIDDING DOCUMENT

Telecommunications Technology Solutions Inc. (TelTech)

Corporate Tender Secretariat · Procurement & Network Engineering Division

RFP-TELTECH-FWA-2026-001 / APPENDIX D

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Site Survey, RF Planning & Civil/Power Engineering Guidelines

TowerCo Co-Location Standard, Greenfield Siting & Link Budget Engineering Standards

DOCUMENT NUMBER	AUTHOR	TARGET SPECTRUM	BIDDING STATUS
RFP-TELTECH-FWA-2026-001	OLIVER TUNGOL	Band n50 (1.5 GHz, 60 MHz TDD)	Official Request for Proposal

1. ENGINEERING PRINCIPLES & SITING ARCHITECTURE

1.1 Siting Philosophy

The primary engineering objective of the 5G FWA rollout is to maximize household population coverage within high-density suburban and rural barangay settlement clusters while minimizing civil capital expenditures and permitting lead times.

* **TowerCo Co-Location First (Option A — Baseline):** The Operator leverages an audited master database of **13,746 operational towers**. For every candidate barangay, field teams must prioritize existing towers within <= 2.0 km (Classes T1, T2, and T3).
* **Greenfield Monopole (Class T4):** Restricted strictly to candidate barangays located > 2.0 km from any verified operational tower structure (~9.3% of sites nationwide; 103 sites in Phase 1).

2. DETAILED TECHNICAL SITE SURVEY REPORT (TSSR) REQUIREMENTS

For every nominal site candidate, the Contractor shall complete and submit a comprehensive TSSR conforming to the following structure:

2.1 General Site Identification & Geography

- Site Name, Candidate ID, Official PSGC Code, Region, Province, Municipality, and Barangay.
- Precision WGS84 GPS Coordinates (Latitude, Longitude recorded with >= 6 decimal places via dual-frequency GPS receiver).
- Ground Elevation Above Mean Sea Level (AMSL in meters).
- Access Road Assessment: Road surface type (concrete, asphalt, macadam, dirt trail), bridge weight limits, and 4x4 hauling accessibility during monsoon seasons.

2.2 Tower Structure & Loading Audit (Co-Location Sites)

- Operating TowerCo Entity (PTCI/MIDC, FTAP Globe, EdgePoint, FTAP Smart, LDIC, FTAP BTS).
- Tower Type (Self-Supporting 3-Leg/4-Leg Tower, Monopole, Guyed Mast, Rooftop Tower).
- Tower Height Above Ground Level (AGL in meters).
- Structural Space Audit: Proposed mounting height for 3x Band n50 antennas (recommended centerline: 30m to 45m AGL).
- Structural Loading Verification: Confirmation that existing tower members and foundation can accommodate additional 3x 8T8R RRUs/AAUs (< 45 kg each) and wind load profile of 250 km/h.

- Panoramic Drone Photographic Survey: High-resolution 360° continuous panoramic video and 12-cardinal-point still photos at proposed antenna centerline height to verify optical line-of-sight across populated settlements.

2.3 Power Infrastructure & Utility Audit

- Existing Commercial AC Utility Power: Electric Cooperative / Distribution Utility name (e.g., MERALCO, PELCO, BENECO, CAPELCO, ILPI).
- Service Entrance Capacity: Transformer rating (kVA), service voltage (230V / 400V, 60 Hz), breaker rating, and phase balance.
- TowerCo DC Power Availability: Sourcing from TowerCo DC busbar (-48V DC) vs. installing Operator-dedicated rectifier subrack.
- Sub-metering feasibility and cable tray routing for power cables.
- Grounding System Audit: Measurement of existing earth ground resistance using a 3-point fall-of-potential earth tester. Earth resistance must be <= 5.0 Ohms.

2.4 Backhaul Transmission Path Survey

- Audited Converge Fiber Network Match (subject to verification on the ground): Distance to nearest confirmed node candidate (Class A <= 0.5 km, Class B <= 1.0 km) or line-tap closure (Class C <= 0.5 km).
- Cable Entry Facility: Manhole, handhole, or aerial pole lead-in routing into tower shelter/compound.
- For Remote Class E/F Sites: Satellite antenna mounting location (unobstructed sky view with 0° to 80° northern elevation arc for Starlink constellation tracking).

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3. RF PROPAGATION & CELL DIMENSIONING GUIDELINES

3.1 Link Budget Parameters (Band n50, 1.5 GHz)

PARAMETER	UNIT	VALUE (DOWNLINK)	VALUE (UPLINK)	TECHNICAL NOTES
Operating Frequency	MHz	1487.0 (Center)	1487.0 (Center)	Band n50 TDD (1457.0 – 1517.0 MHz Channel Range)
Channel Bandwidth	MHz	60.0	60.0	162 Resource Blocks (SCS 30 kHz)
BTS Maximum Conducted Power	dBm / W	55.0 dBm (320 W)	N/A	8 x 40W RRU configuration
BTS Power per Resource Block	dBm	32.9 dBm	N/A	Uniform power distribution
BTS Antenna Gain	dBi	17.5 dBi	17.5 dBi	8-port cross-pol macro antenna
BTS Cable / Jumper Loss	dB	1.0 dB	1.0 dB	Short optical/RF jumper interface
CPE Maximum Transmit Power	dBm / W	N/A	23.0 dBm (200 mW)	Power Class 3 UE (or Class 2 26 dBm)
CPE Antenna Gain	dBi	9.0 dBi	9.0 dBi	High-gain directional indoor/outdoor
CPE Body / Indoor Penetration Loss	dB	8.0 dB	8.0 dB	Concrete hollow block & glass attenuation
Foliage / Clutter Margin	dB	6.0 dB	6.0 dB	Typical rural tropical tree canopy
Shadowing Margin (Log-Normal)	dB	7.0 dB	7.0 dB	90% edge cell coverage reliability
Thermal Noise Density	dBm/Hz	-174.0	-174.0	kT @ 290K
Receiver Noise Figure	dB	6.0 dB (CPE)	3.5 dB (gNodeB)	Active RF front-end
Target SINR for Min Service	dB	-1.0 dB	-2.0 dB	QPSK rate 1/3 (Cell Edge 25 Mbps DL)

PARAMETER	UNIT	VALUE (DOWNLINK)	VALUE (UPLINK)	TECHNICAL NOTES
Maximum Allowable Path Loss (MAPL)	dB	144.5 dB	139.8 dB	Uplink is coverage limiting
Calculated Reliable Cell Radius	km	{3.5 km}	{3.2 km}	Standard suburban/rural clutter

3.2 Nominal Sector Azimuth & Tilt Strategy

- **Sector Count:** 3 sectors per macro site with 120° nominal separation (0°, 120°, 240°). Azimuths must be tailored during TSSR to track actual barangay population concentrations along transport corridors and coastal plains.
- **Electrical Remote Downtilt (eRET):** Antennas must feature AISG 2.0 remote tilt adjustable between 2° and 10° from the central OSS. Mechanical tilt shall only be applied when required to clear close-in obstructions (>15°).

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4. CIVIL WORKS & STRUCTURAL ENGINEERING (CLASS T4 GREENFIELD SITES)

For the ~10% of sites requiring self-constructed greenfield towers, the Contractor shall adhere to the following civil standards:

1. **Monopole Tower Specification:**

* 30-meter or 40-meter polygonal tubular steel monopole constructed in 3 to 4 slip-joint sections.
* Structural steel: ASTM A572 Grade 50 (or equivalent high-strength structural grade) hot-dip galvanized per ASTM A123 (>= 85microns zinc coating).
* Design Wind Speed: Minimum **250 km/h 3-second gust** in accordance with the National Structural Code of the Philippines (NSCP 2015 / TIA-222-G/H).
* Top antenna mounting platform capable of supporting 6x antennas/RRUs and microwave dish mountings.

2. **Compound Civil Works:**

* Compound Footprint: Standard 8.0m x 8.0m (64 m²) gravel-topped compound.
* Perimeter Fencing: 2.4-meter high cyclone wire mesh fence topped with 3 strands of galvanized concertina razor wire.
* Equipment Pad: Reinforced concrete equipment plinth (Class A concrete, 28-day compressive strength >= 21MPa / 3,000psi) for outdoor cabinets and generator set.

3. **Lightning Protection & Grounding Grid:**

* Air terminal (lightning rod) installed at monopole top extending >= 2.0m above uppermost antenna.
* Down-conductor: 70mm² bare stranded copper cable routed externally and clamped every 1.0m to tower shaft.
* Earth ground ring: Continuous buried bare copper loop around tower base and compound with copper-clad steel ground rods (19mm dia x 3.0m length) exothermically welded (Cadweld) at all junctions, achieving <= 5.0 Ohms.

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5. SINGLE SITE VERIFICATION (SSV) & CLUSTER ACCEPTANCE CRITERIA

Before any site is granted Commercial On-Air Acceptance, the Contractor must complete and submit a signed SSV Report verifying:

1. **Stationary Functional Tests:**

* Sector Verification: Physical azimuth and tilt match TSSR approval within +/- 2°.
* RF Transmit Power: Conducted power on all 8 transmit paths within +/- 0.5 dB of nominal.
* VSWR / Return Loss: Return loss >= 15 dB across all antenna feeder lines.
* Max Throughput Test: Single CPE situated in line-of-sight achieving >= 450 Mbps DL and >= 60 Mbps UL.

2. **Short Drive Test & Static Multi-Point Test:**

* 5 stationary measurement points per sector (Near: 500m, Mid: 1.5 km, Far/Edge: 3.0 km).
* RSRP >= -95 dBm across >= 90% of surveyed barangay inhabited zone.
* Ping RTT <= 20 ms to local UPF IP address; zero packet loss over 1,000 continuous pings.
* Inter-sector handover execution without drop during transition.

(End of Appendix D — Site Survey, RF Planning & Civil Engineering Guidelines)